

Level Up:

Choosing the Technical Leadership Path

pat@patkua.com

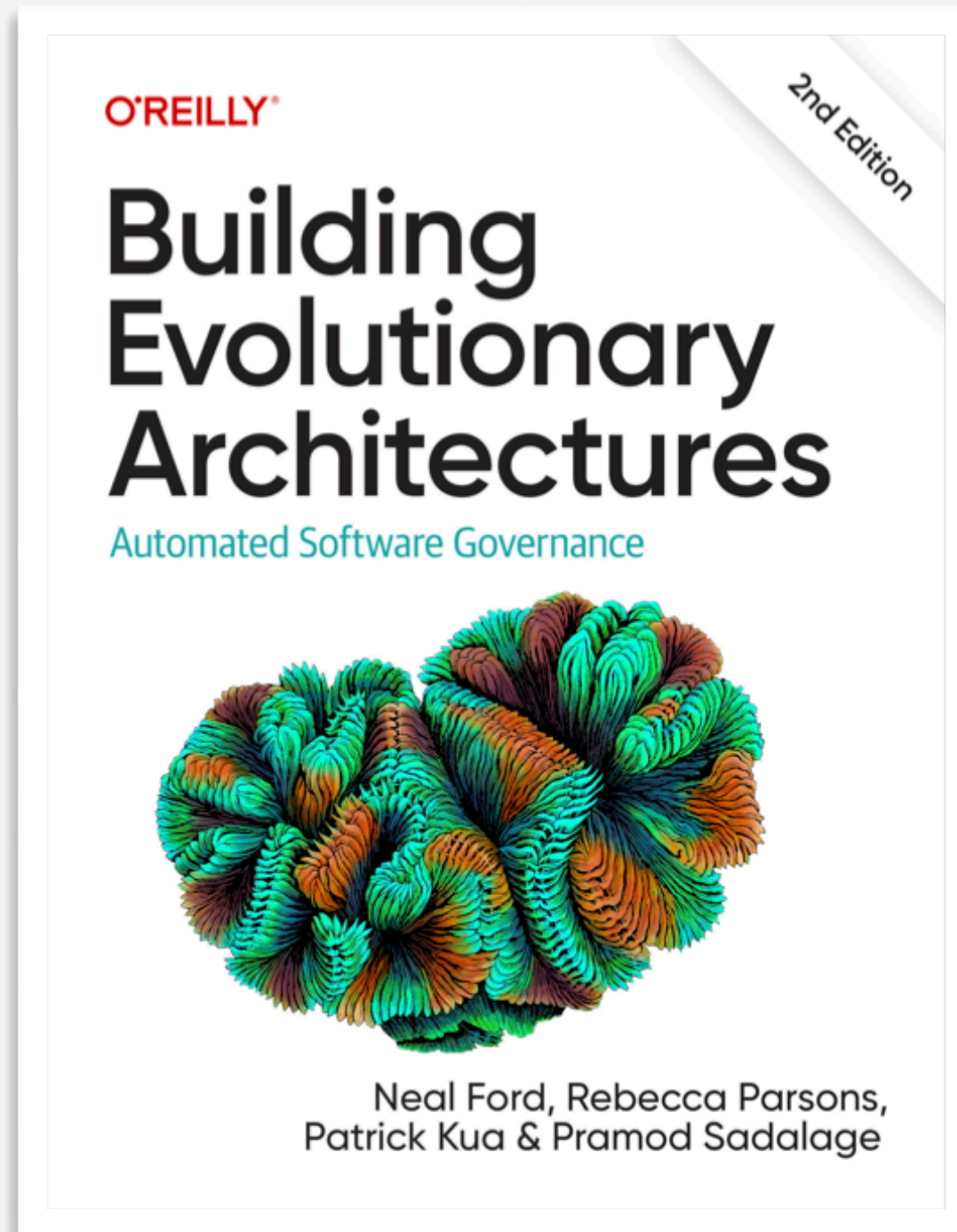


TechLead
ACADEMY

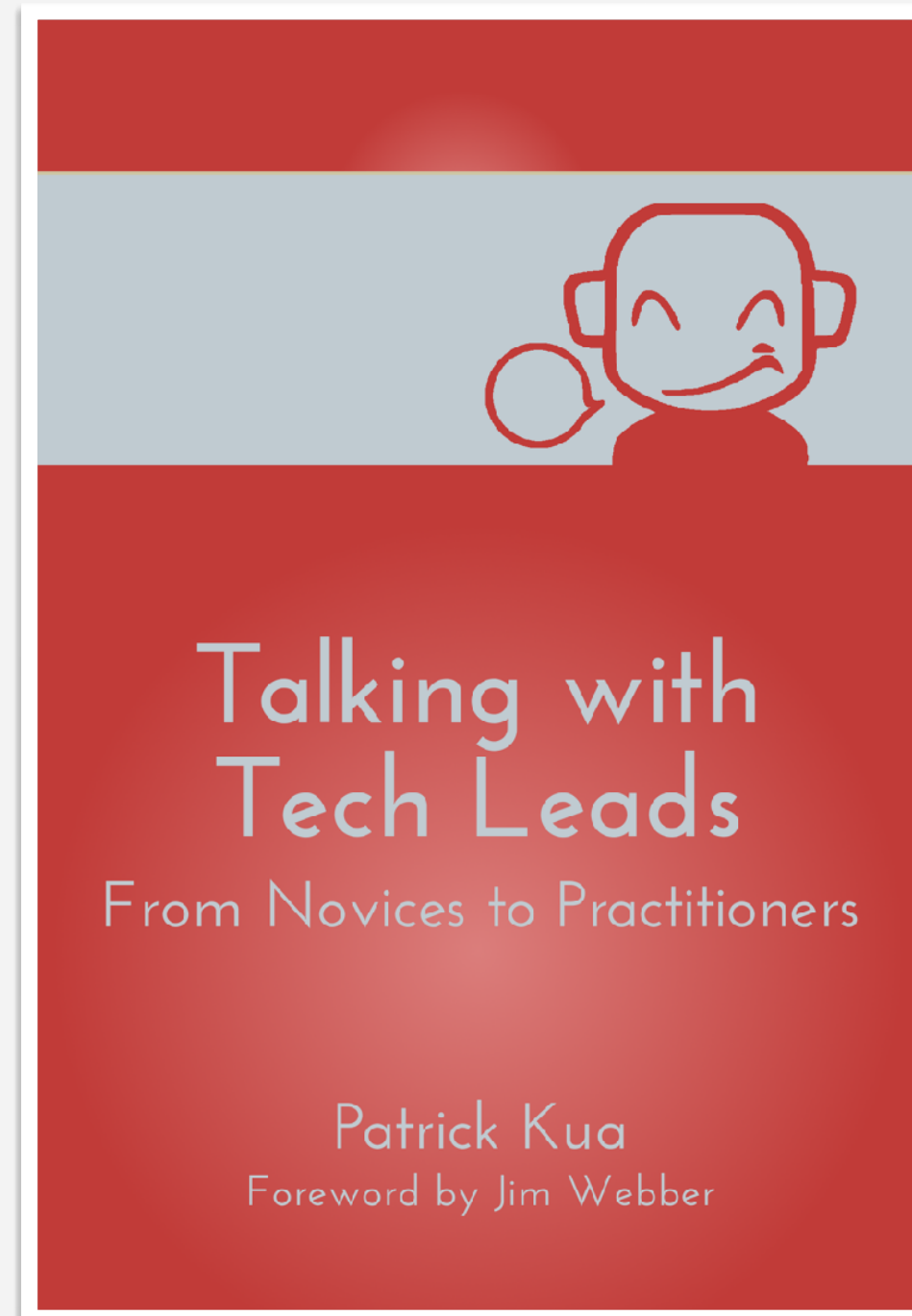
@patkua



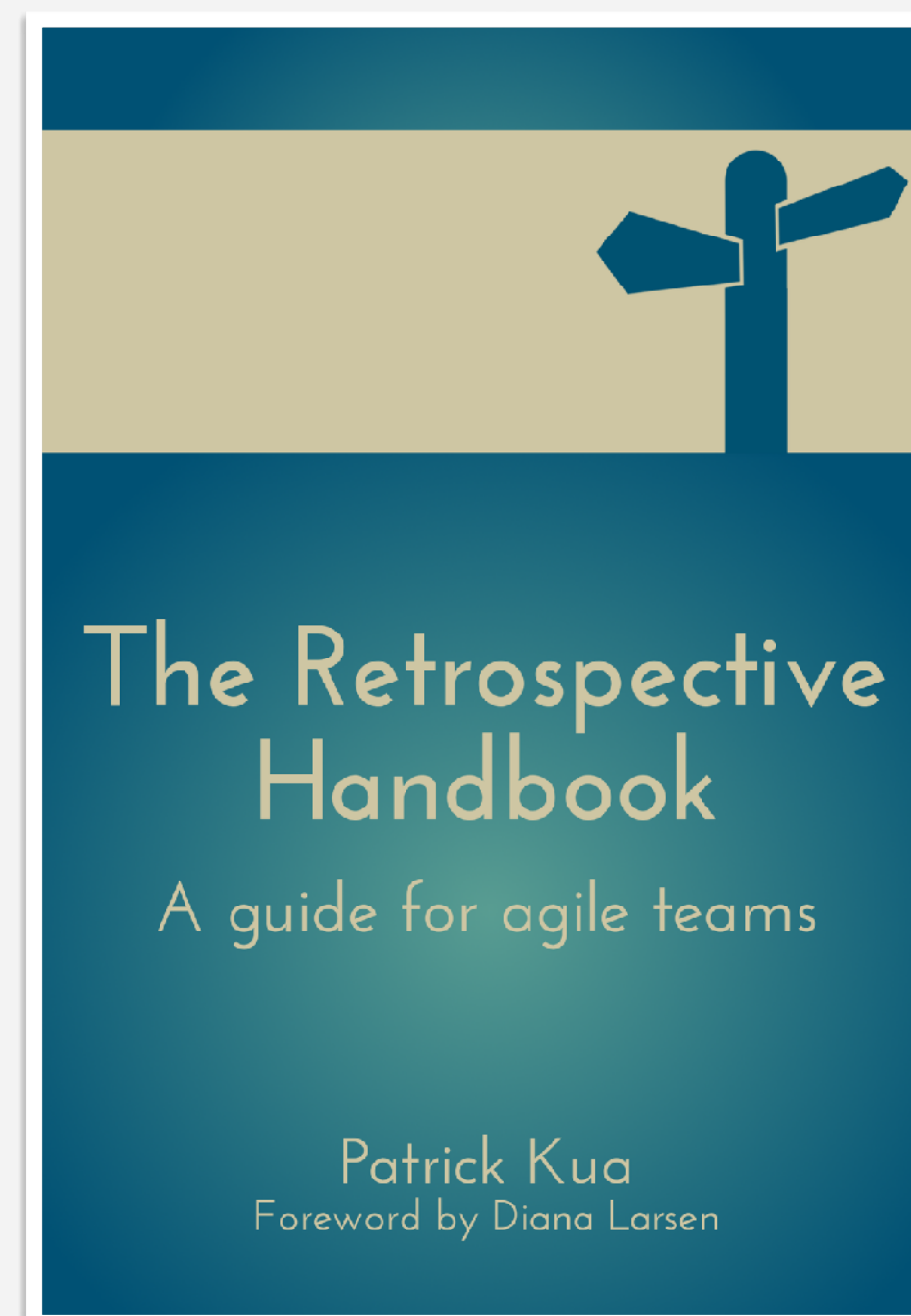
~25 years experience



thekua.io/evolarch



thekua.io/twtl



thekua.io/retrobook

#Architect #Developer #Coach #Leader #CTO

#Life-long learner #Author #Speaker

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TechLead

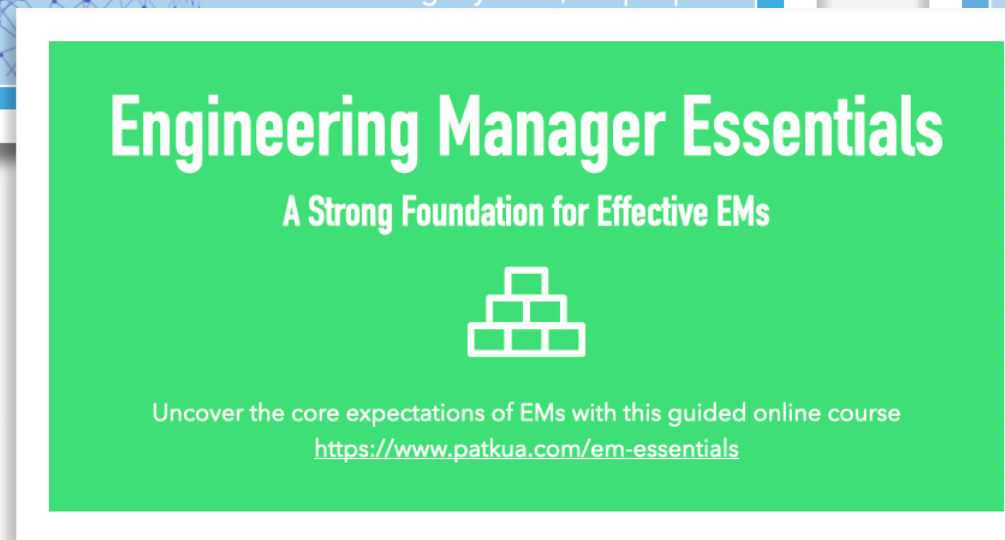
ACADEMY

10+ Years Training

e.g. TLs, EMs, Staff+

Coaching

e.g. CTOs, VPs of Eng



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Exploring Technical Leadership



Reasons For You



How to Get Started



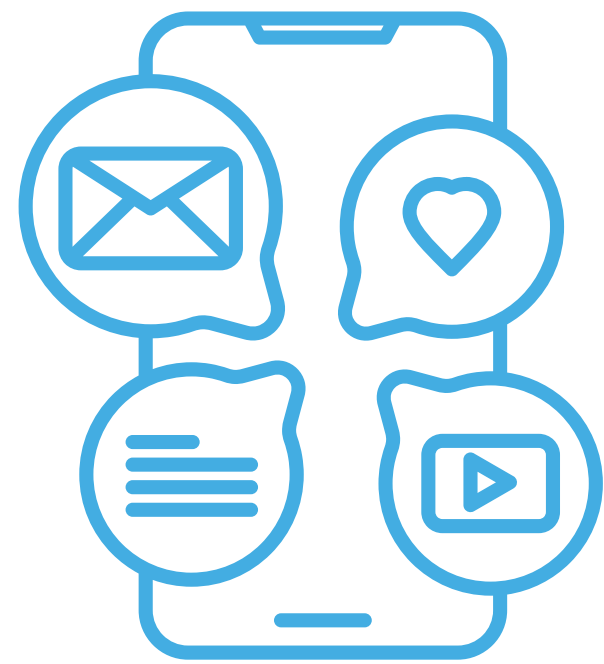
Exploring Technical Leadership





STORY

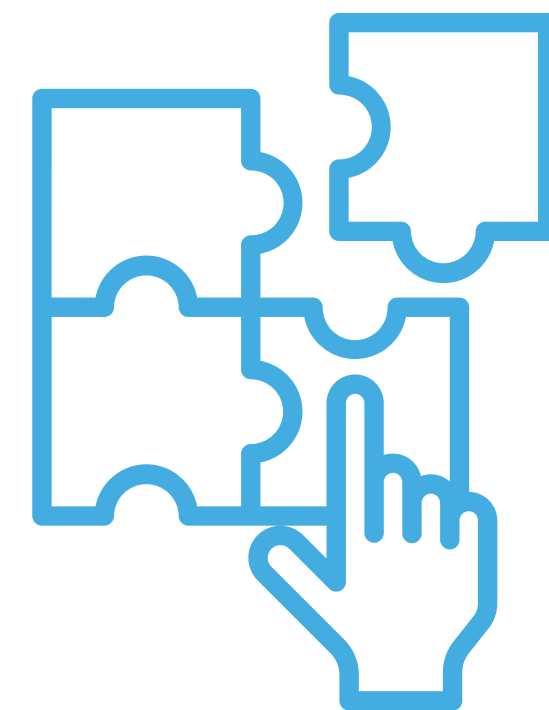
Story



Social Media
App



Buying
spree



8 different
Companies



"Just" delete
unwanted
components

Major delays



Low # phones supported

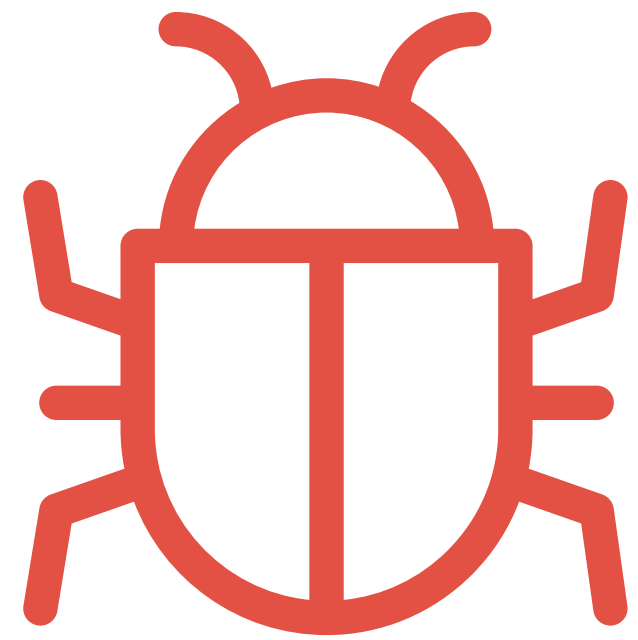


Missing differentiators

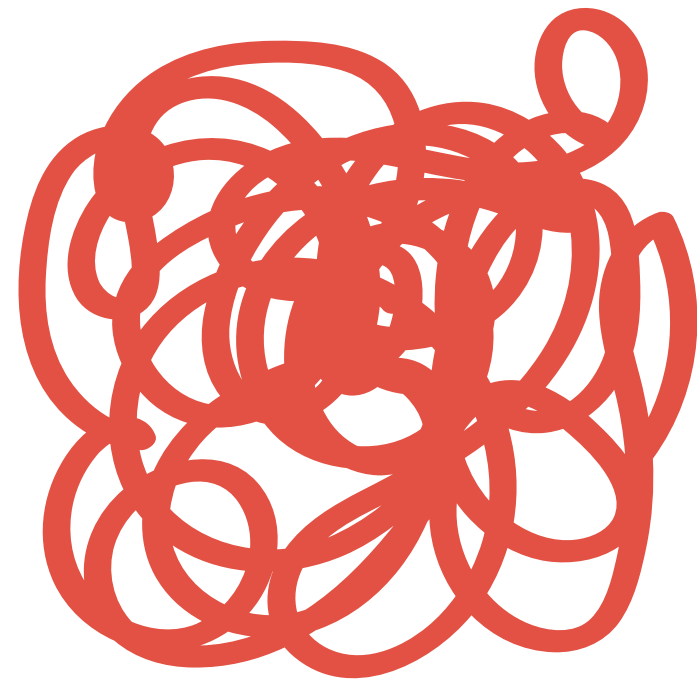


The result...

(Lack of) Technical Alignment



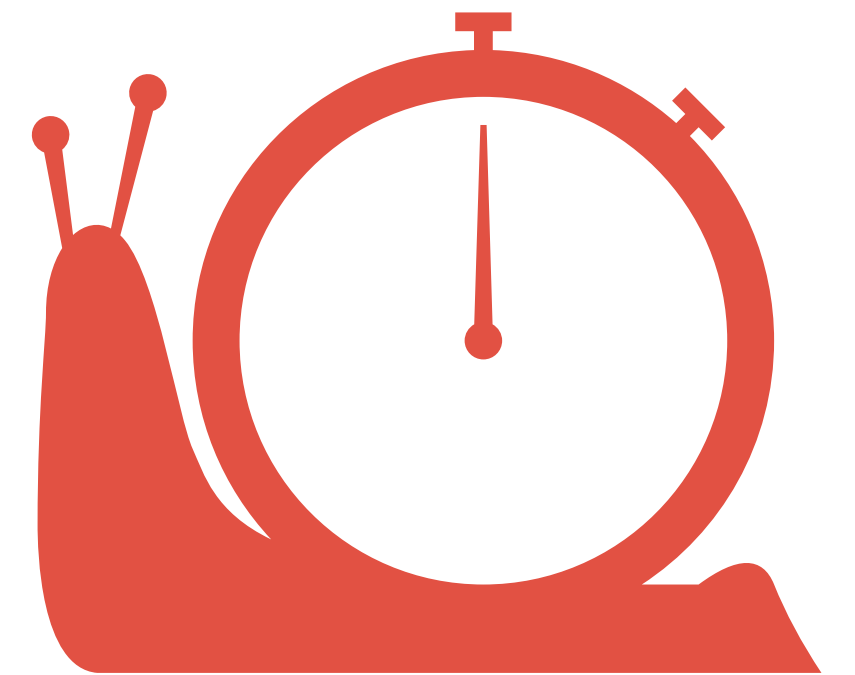
Bugs



↑ Complexity



Bad UX



↓ Progress

"I have to implement this new feature."

*"Why do we have ten(!!!) different ways
to raise events? 🤔"*

"This is going to make my work a lot harder 😡"

No Silver Bullet

Essence and Accidents of Software Engineering

Frederick P. Brooks, Jr.
University of North Carolina at Chapel Hill

Fashioning complex conceptual constructs is the essence; accidental tasks arise in representing the constructs in language. Past progress has so reduced the accidental tasks that future progress now depends upon addressing the essence.

Of all the monsters that fill the nightmares of our folklore, none terrify more than werewolves, because they transform unexpectedly from the familiar into horrors. For these, one seeks bullets of silver that can magically lay them to rest.

The familiar software project, at least as seen by the nontechnical manager, has something of this character; it is usually innocent and straightforward, but is capable of becoming a monster of missed schedules, blown budgets, and flawed products. So we hear desperate cries for a silver bullet—something to make software costs drop as rapidly as computer hardware costs do.

But, as we look to the horizon of a decade hence, we see no silver bullet. There is no single development, in either technology or in management technique, that by itself promises even one order-of-magnitude improvement in productivity, in reliability, in simplicity. In this article, I shall try to show why, by examining both the nature of the software problem and the properties of the bullets proposed.

Skepticism is not pessimism, however. Although we see no startling break-

throughs—and indeed, I believe such to be inconsistent with the nature of software—many encouraging innovations are under way. A disciplined, consistent effort to develop, propagate, and exploit these innovations should indeed yield an order-of-magnitude improvement. There is no royal road, but there is a road.

The first step toward the management of disease was replacement of demon theories and humours theories by the germ theory. That very step, the beginning of hope, in itself dashed all hopes of magical solutions. It told workers that progress would be made stepwise, at great effort, and that a persistent, unremitting care would have to be paid to a discipline of cleanliness. So it is with software engineering today.

Does it have to be hard?—Essential difficulties

Not only are there no silver bullets now in view, the very nature of software makes it unlikely that there will be any—no inventions that will do for software productivity, reliability, and simplicity what electronics, transistors, and large-scale integration did for computer hardware.

This article was first published in *Information Processing '86*, ISBN No. 0-444-70077-3, H.-J. Kugler, ed., Elsevier Science Publishers B.V. (North-Holland) © IFIP 1986.

COMPUTER

Essential complexity Accidental complexity

1987

@patkua

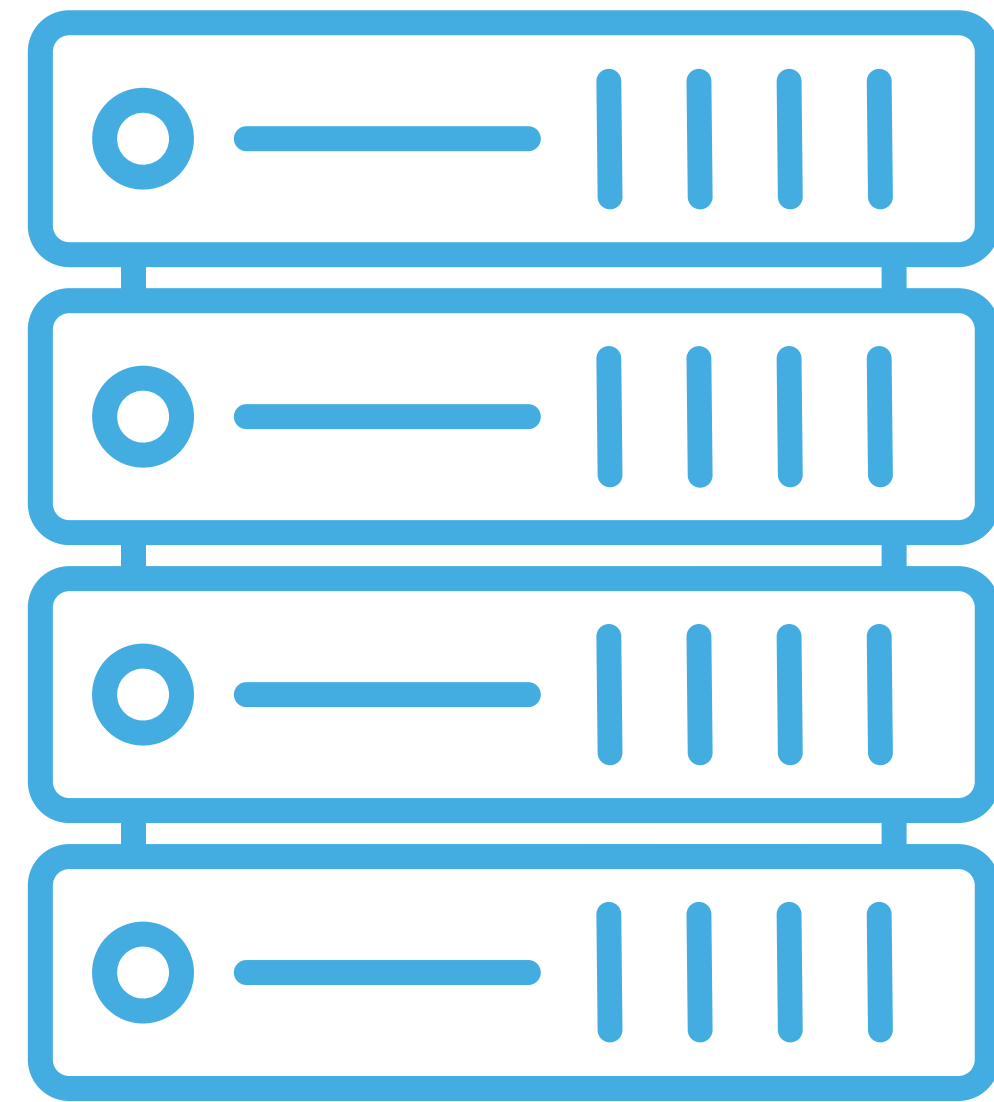
Accidental complexity

THE
C
PROGRAMMING
LANGUAGE

VS



Accidental complexity



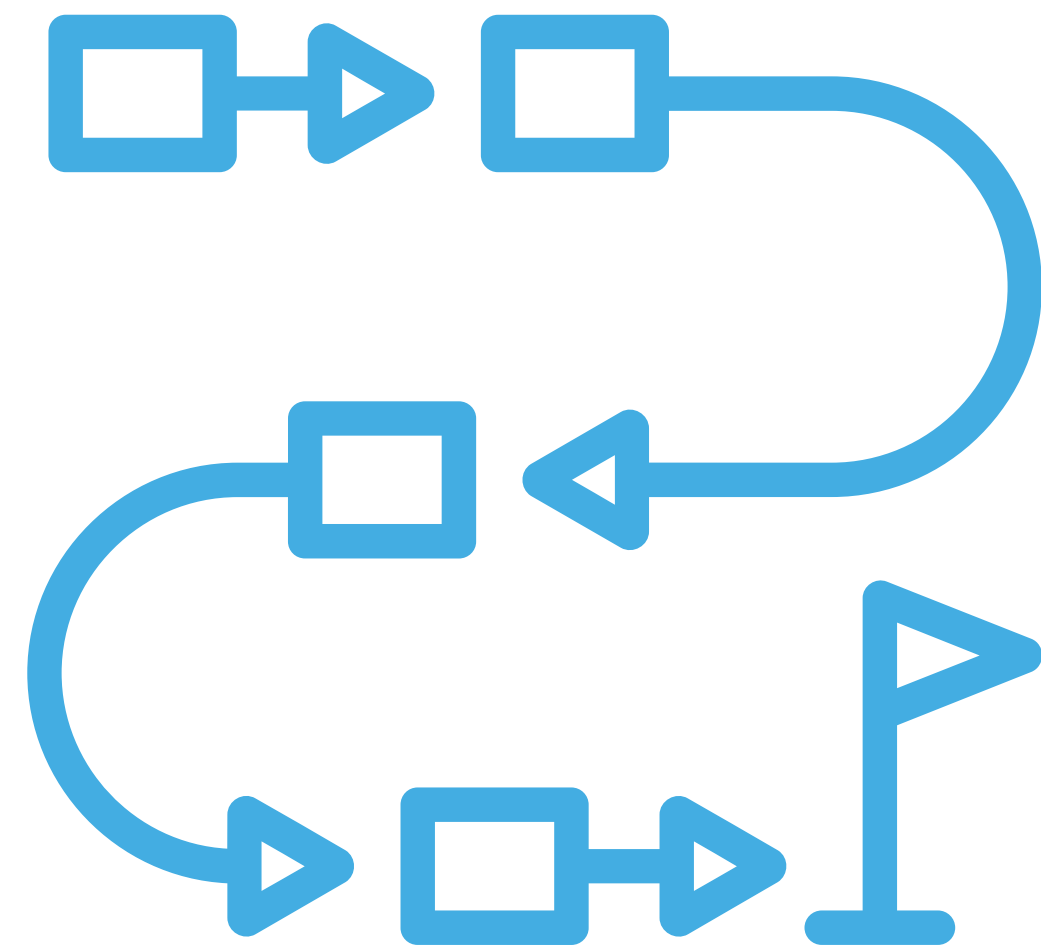
Self-hosted

VS

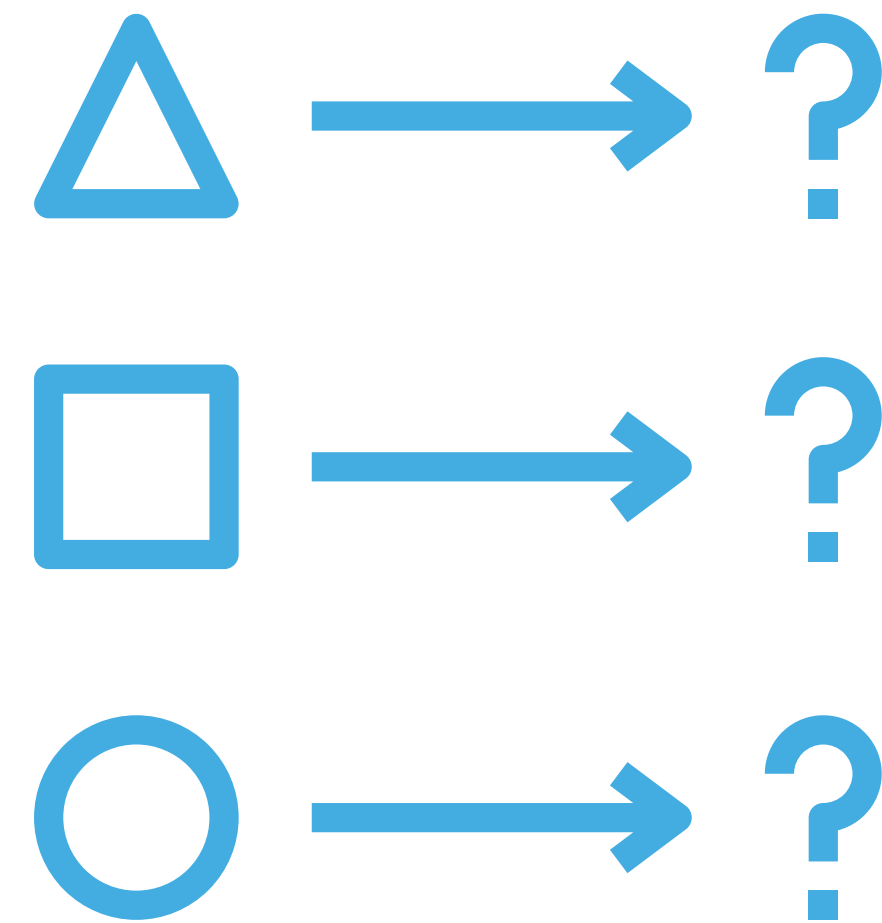


Cloud

Accidental complexity



vs



Inappropriate
abstractions

"Simple" code



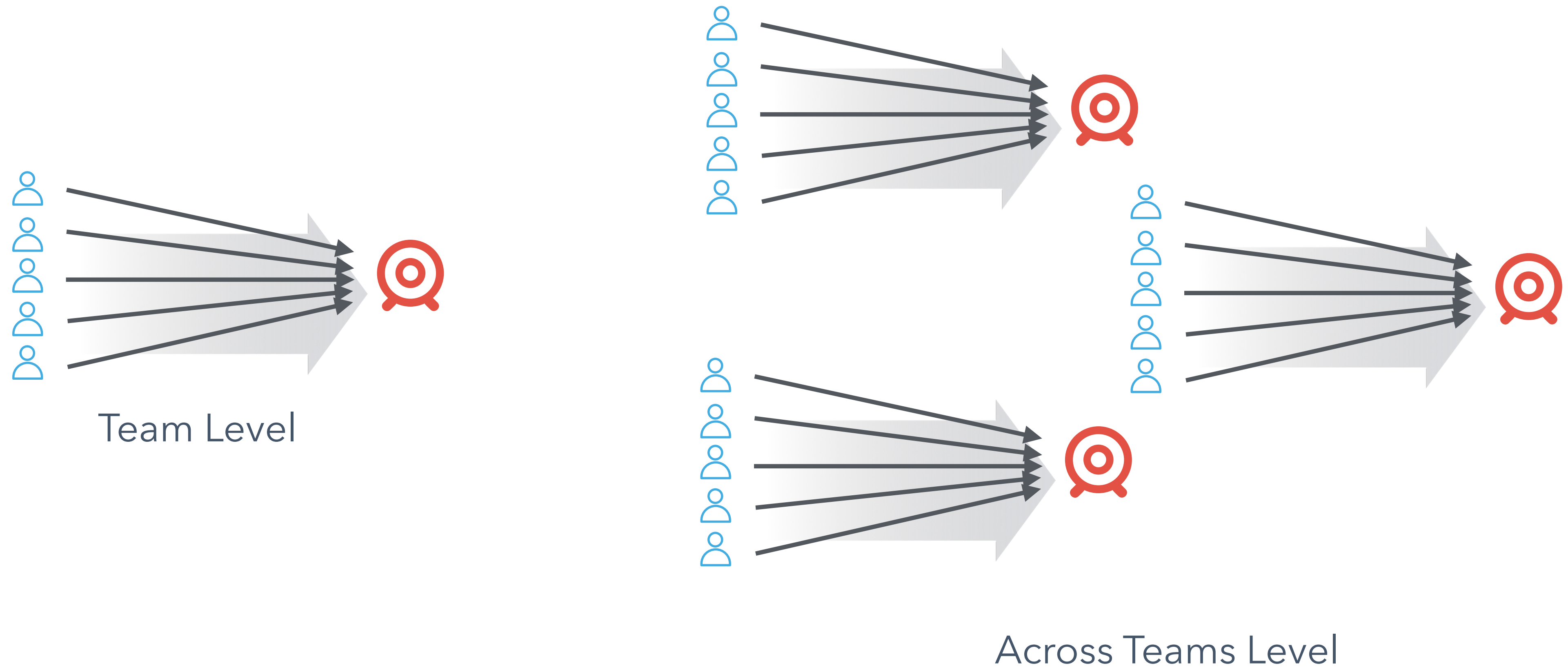
“A lack of technical leadership increases accidental complexity”

What is Technical Leadership?

Technical Leadership is...

*“The **act** of aligning a group of people in a technical context”*

Technical Alignment



Example: Technical Alignment

“Let’s see how we implement email notifications in this system 🧐”

instead of

“This will be a fun way to implement this email notification... 😎”

Example: Technical Alignment

*“We have different approaches to error handling.
Let’s agree on our standard approach”*

instead of

*“I’ll handle this error later. I’ll propagate it
for now”*

Technical Alignment

(Lack of)

- ▶ Group of individuals
- ▶ Prioritises individual task
- ▶ “I’ll do it my own way”

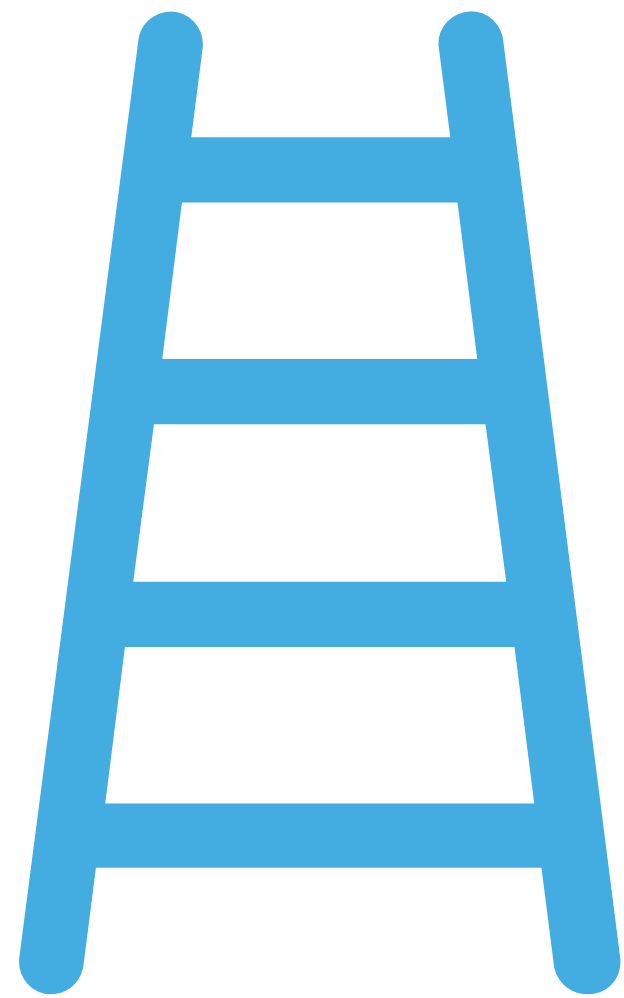
(Stronger)

- ▶ High performing team
- ▶ Prioritises long-term
- ▶ Strong collaboration

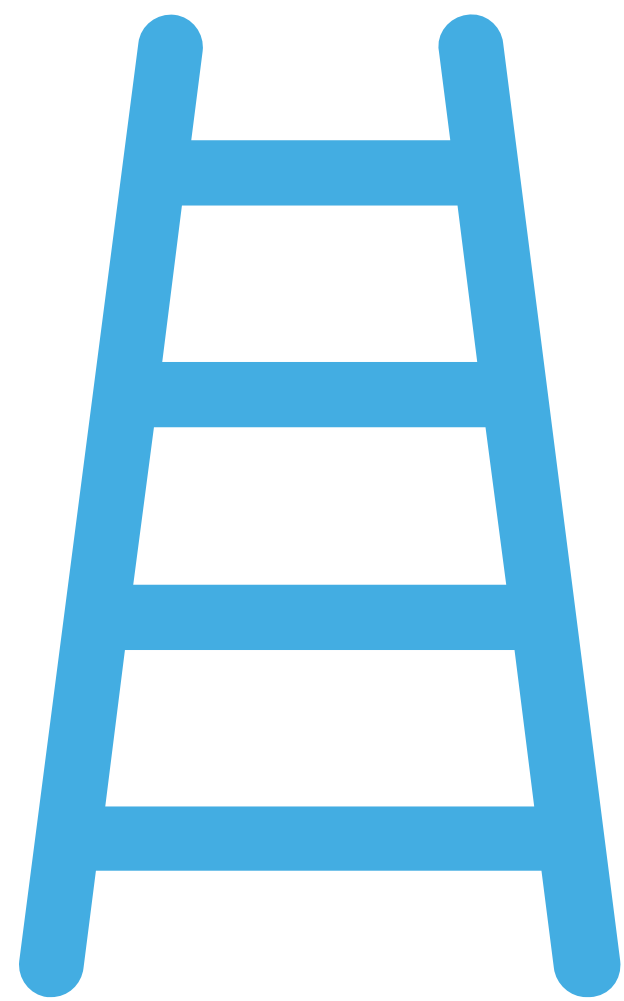
The background of the image is a festive party scene. It shows a crowd of people, some wearing party hats, with a large amount of multi-colored confetti (pink, blue, yellow, green) falling through the air. The overall lighting is bright and celebratory. A semi-transparent white rectangular box is overlaid on the upper portion of the image, containing the main text.

"An investment in the right level of technical alignment makes future change easier"

Who is accountable for technical leadership?



Role	Behaviour	Impact
Position	X	X
Position	X	X
Position	X	X
Position	X	X



Tech Lead

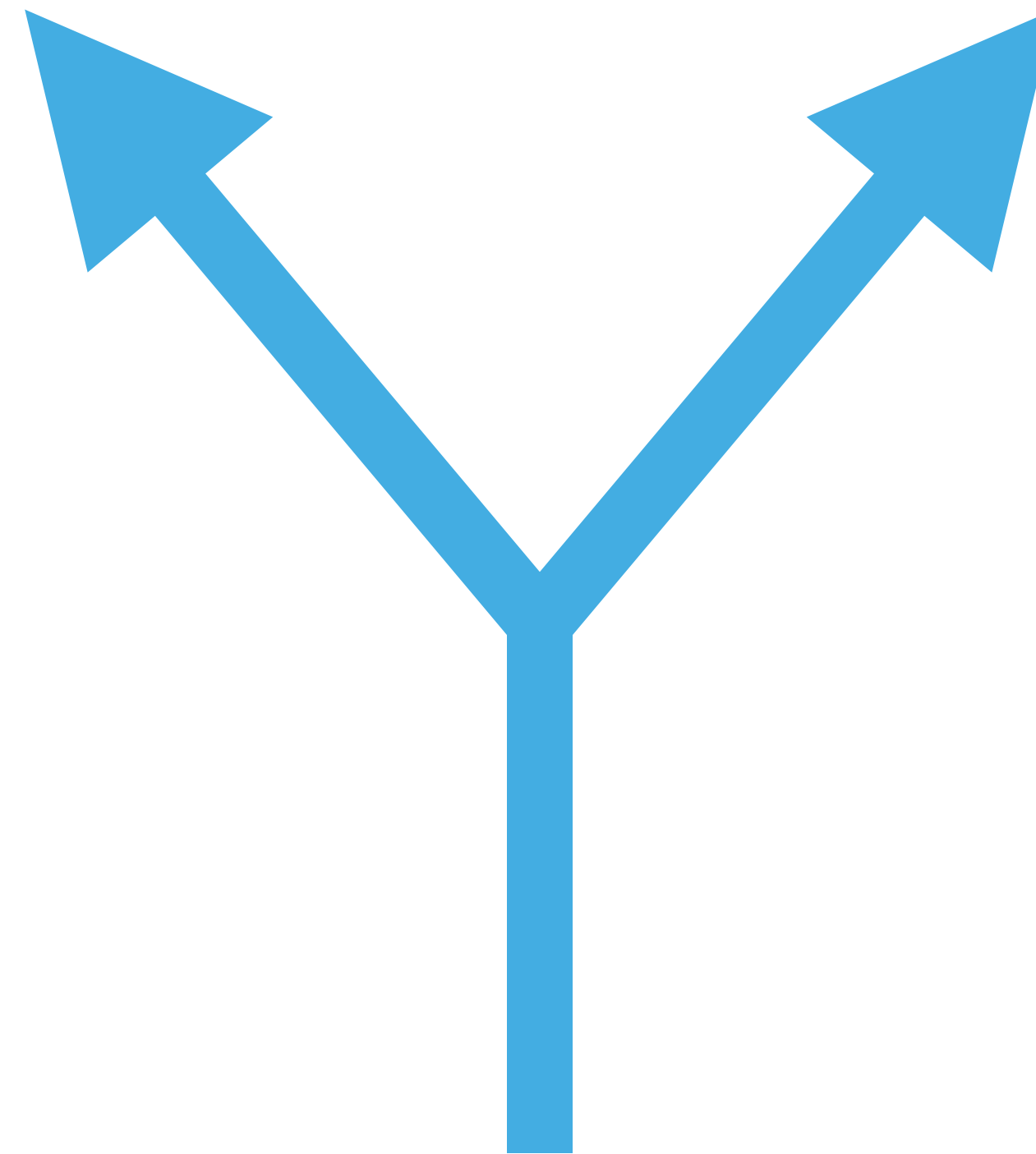
Senior Software Engineer

Software Engineer

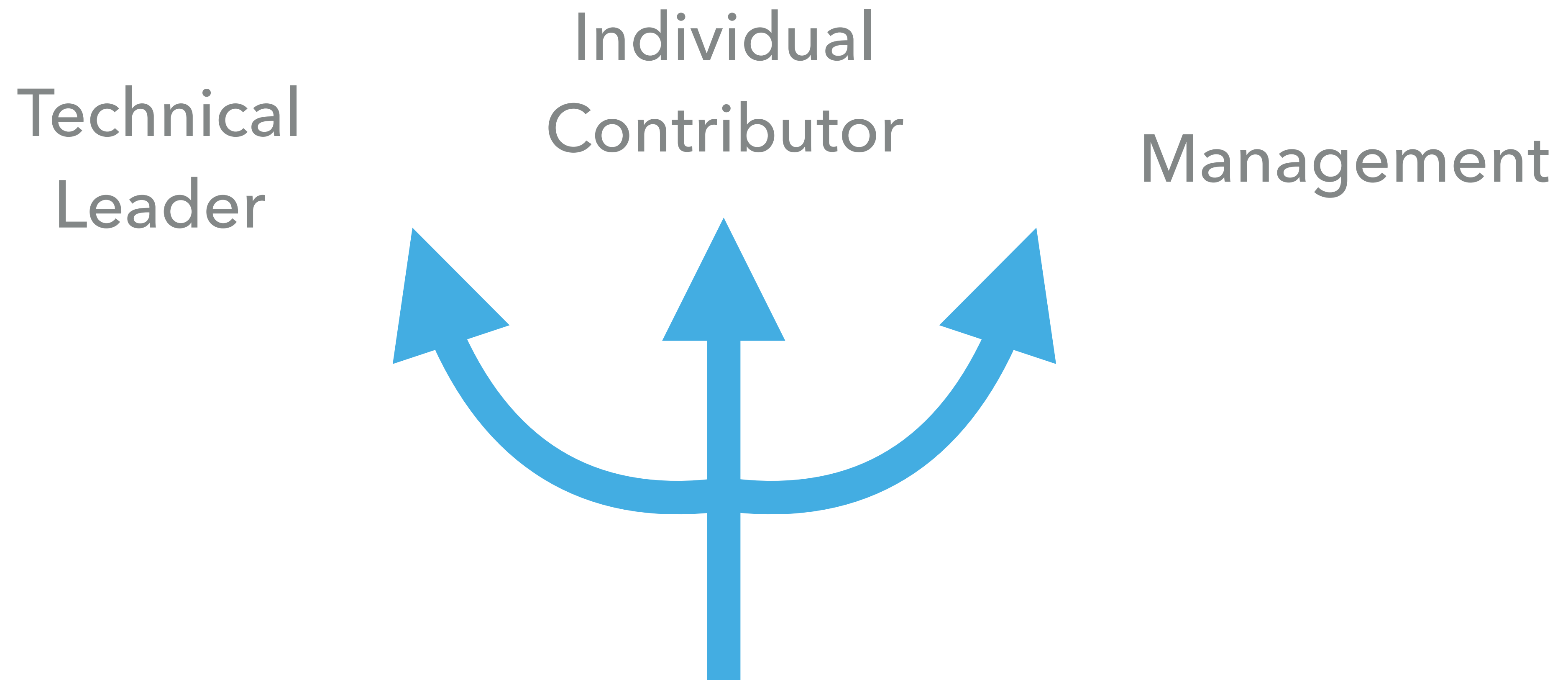
Junior Software Engineer

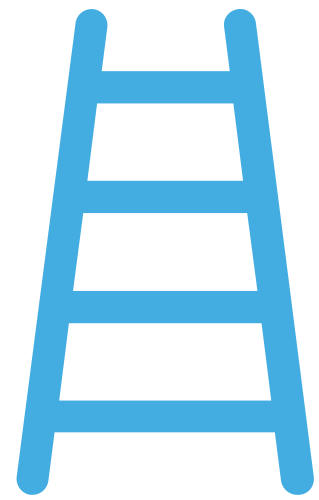
Individual
Contributor

Management



Trident Model





Technical Leader

Lead Dev
Tech Lead
Architect
Staff Engineer*

Individual Contributor

Staff Engineer*

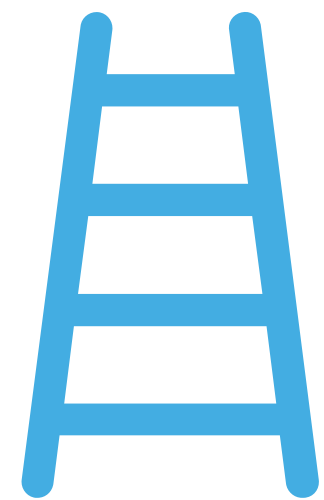
Senior Software
Engineer

Software Engineer

Junior
Software Engineer

Manager

Engineering
Manager



Technical Leader

Lead Dev
Tech Lead
Architect
Staff Engineer*

Individual Contributor

Staff Engineer*

Senior Software
Engineer

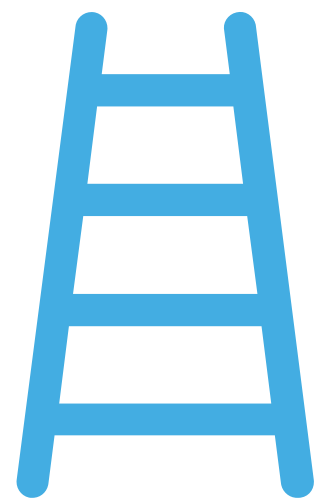
Software Engineer

Junior
Software Engineer

Manager

Engineering
Manager





Technical Leader

Lead Dev
Tech Lead
Architect
Staff Engineer*

Individual Contributor

Staff Engineer*

Senior Software
Engineer

Software Engineer

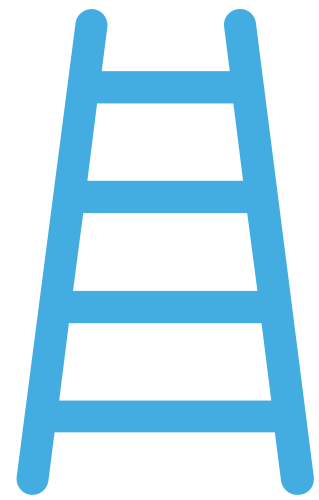
Junior
Software Engineer

Manager

Engineering
Manager

← Create some

Accountable



Technical Leader

Lead Dev
Tech Lead
Architect
Staff Engineer*

Individual Contributor

Staff Engineer*

Senior Software
Engineer

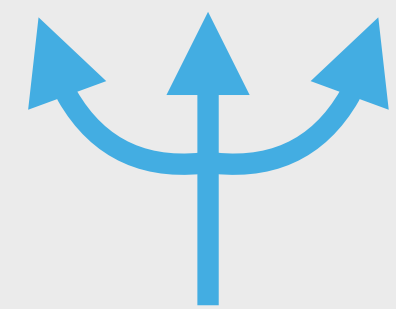
Software Engineer

Junior
Software Engineer

Manager

Engineering
Manager

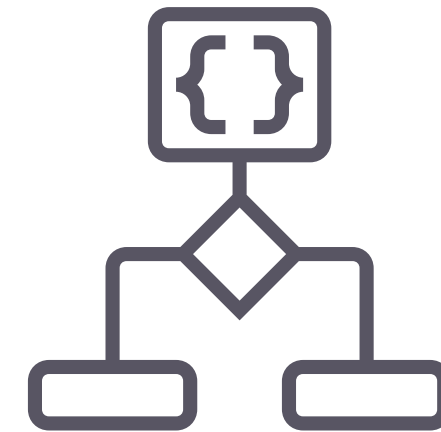
Individual Contributor



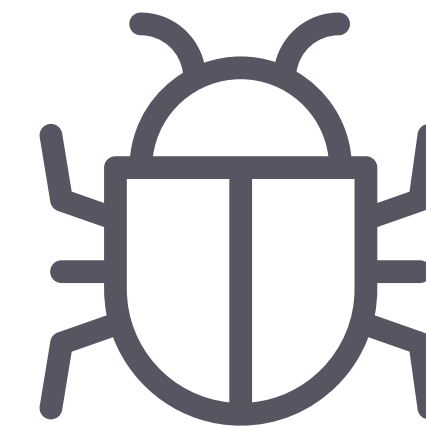
**Trident
Model**

70-80% time spent

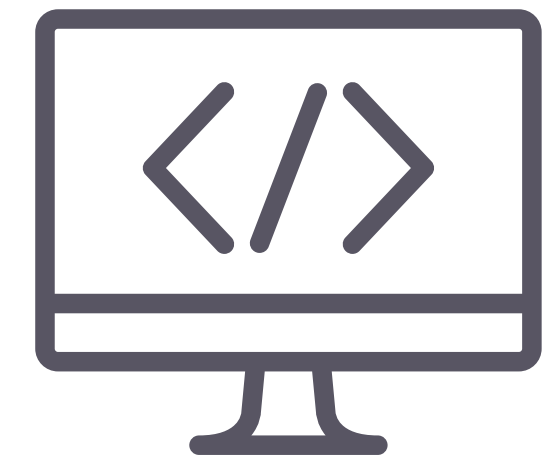
“Executing, doing”



Designing



Testing

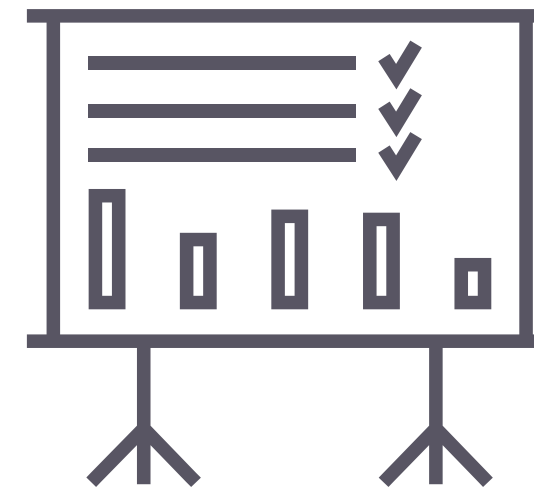


Coding

Management

70-80% time spent

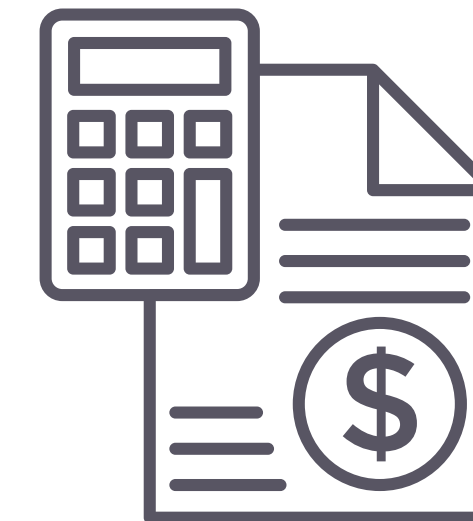
“Managing the system”



Planning



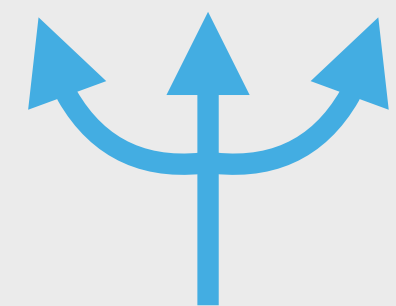
Organising



Budgeting



Supporting

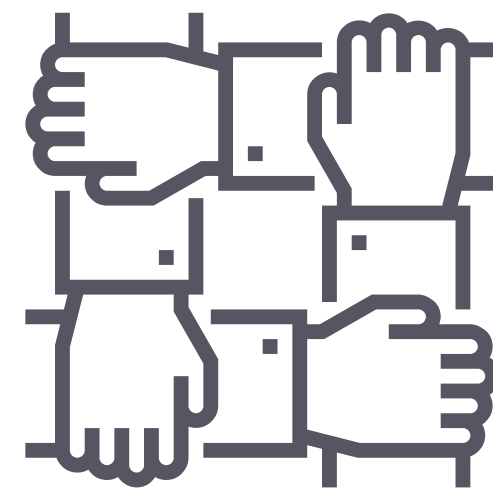


**Trident
Model**

Technical Leader

70-80% time spent

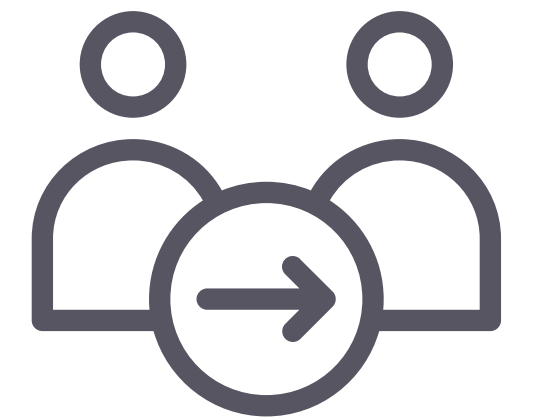
“Leading technical topics and teams”



Aligning Team



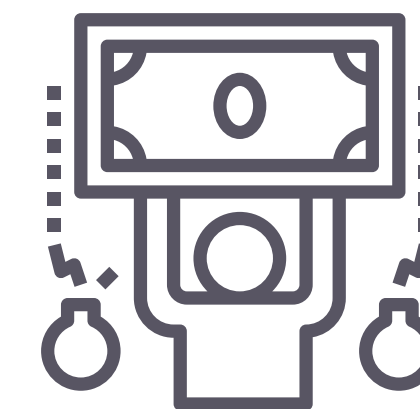
Technical Vision



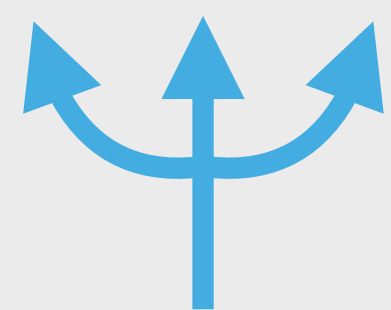
Growing
Technical
Knowledge



Technical Risk
Management



Tech Debt
Management



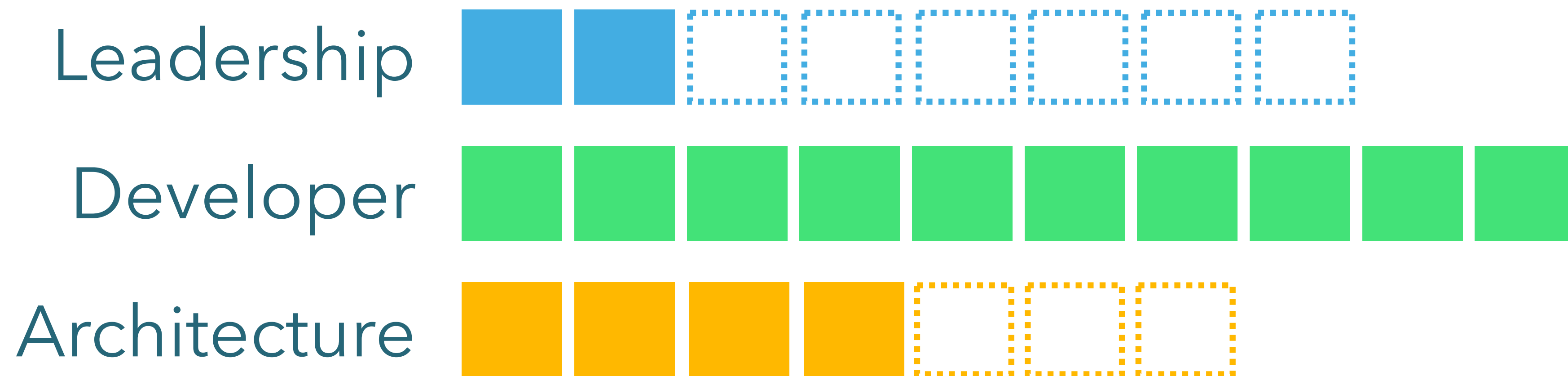
**Trident
Model**



*"What got you here won't
get you there"*

"Invest in leadership skills"

Skill Distribution



Leadership Skills

Empathy

Self-Awareness

Motivation

Conflict Resolution

Communication

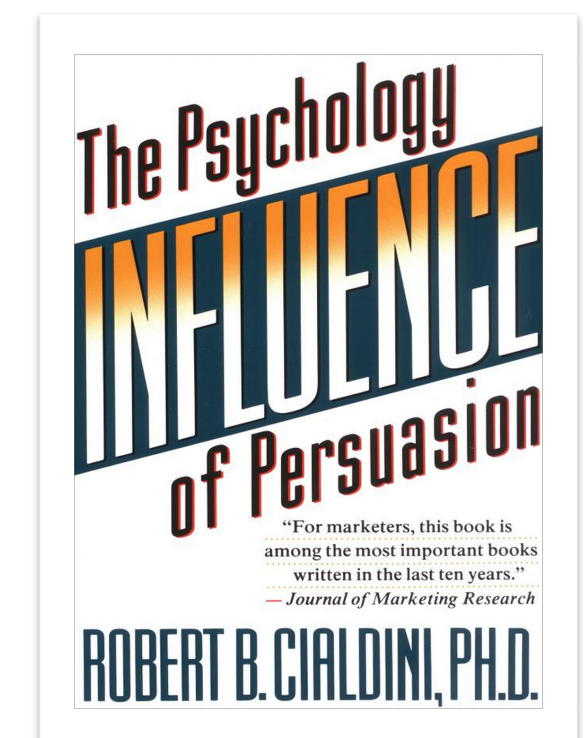
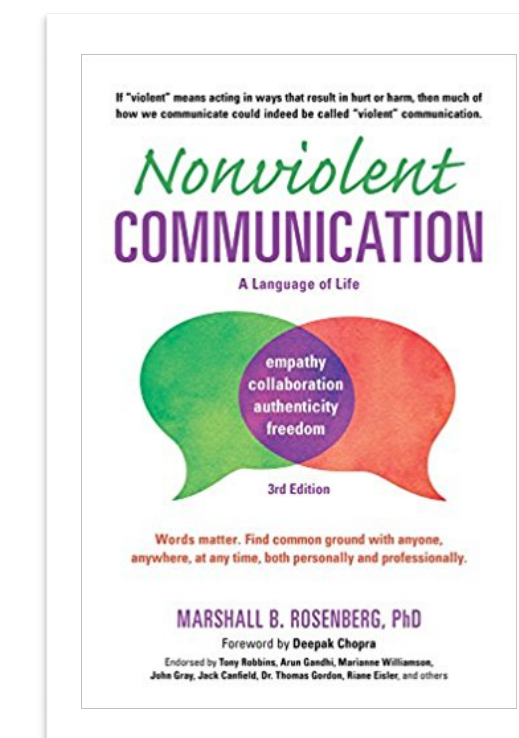
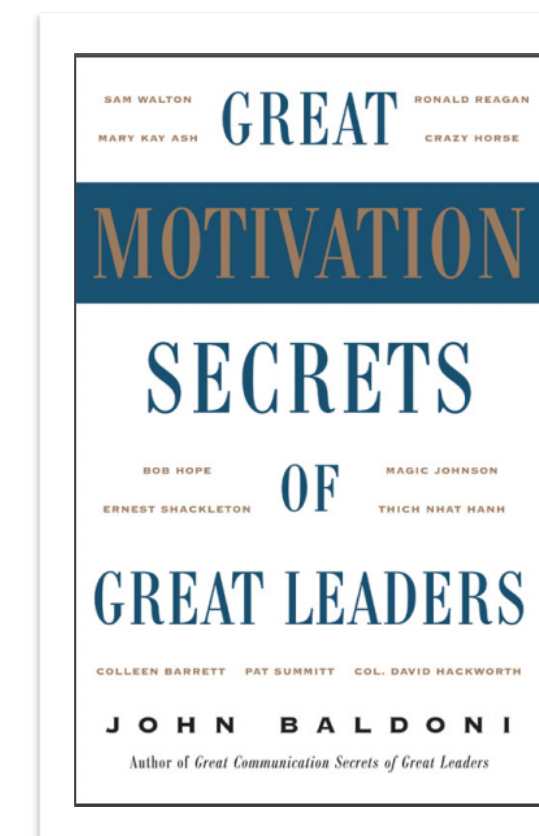
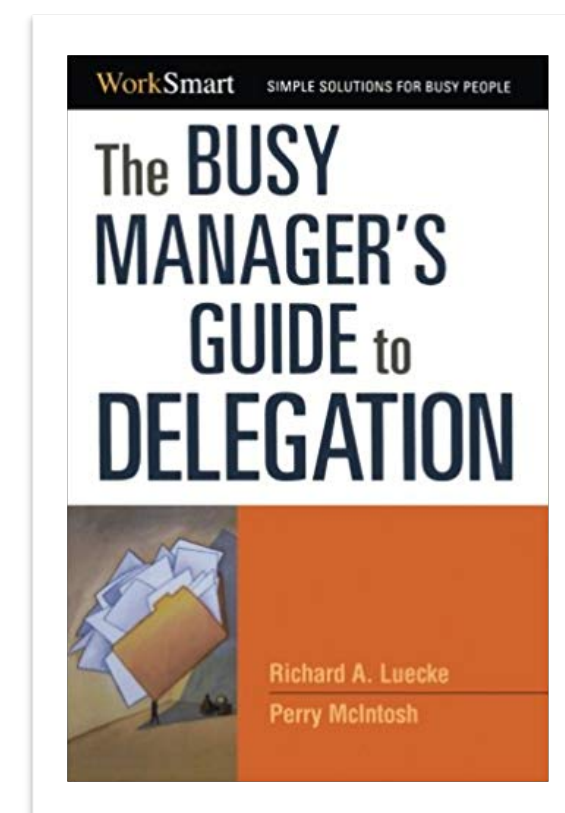
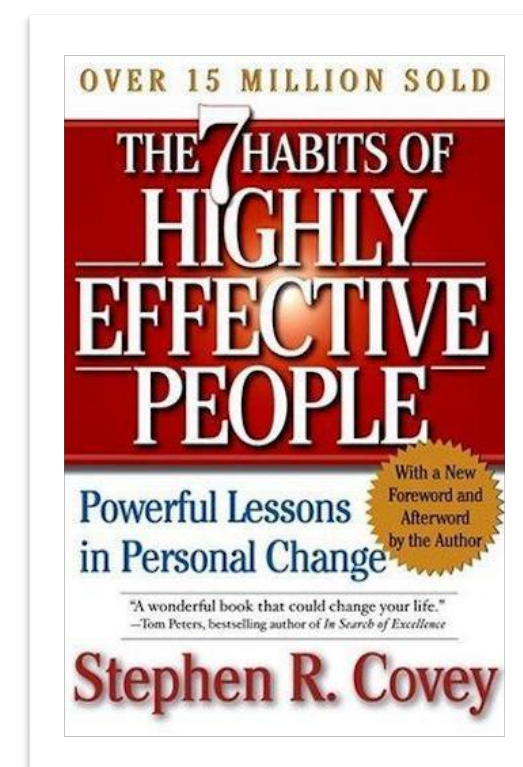
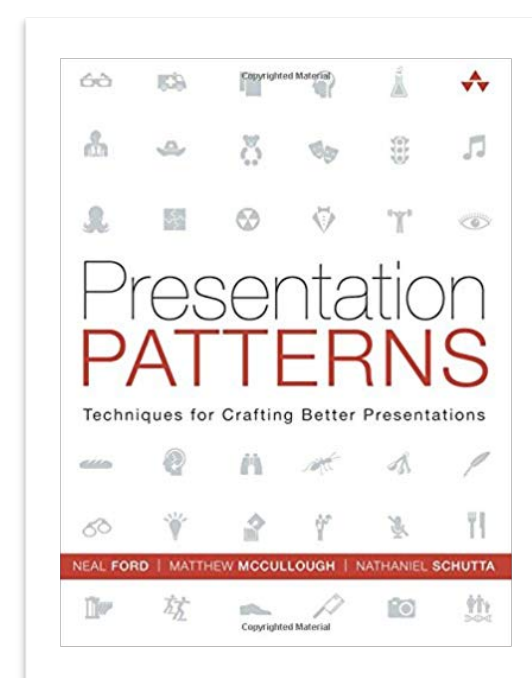
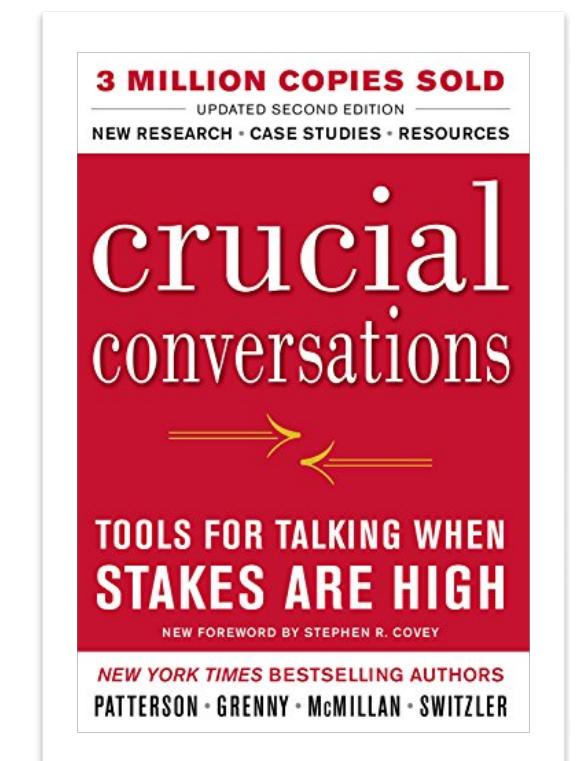
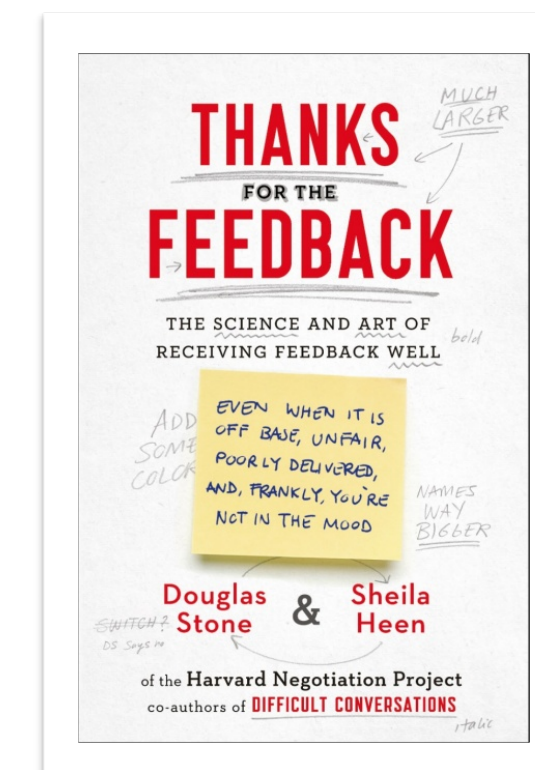
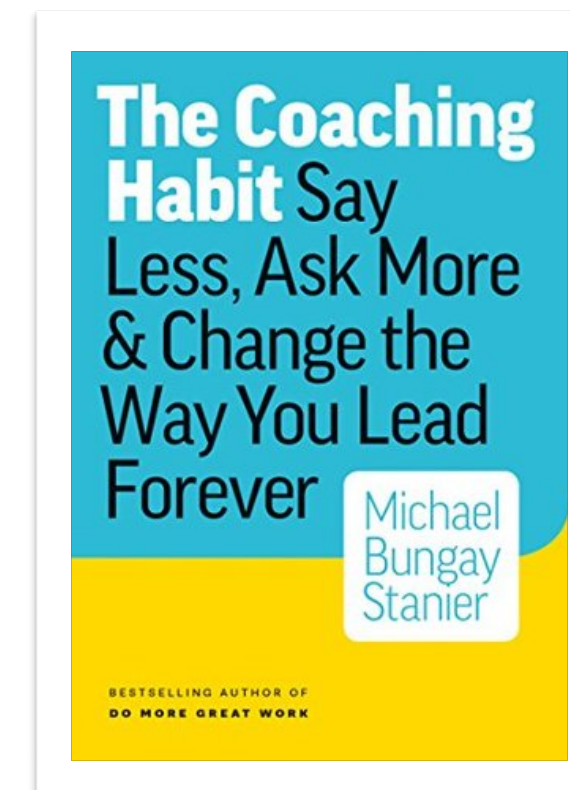
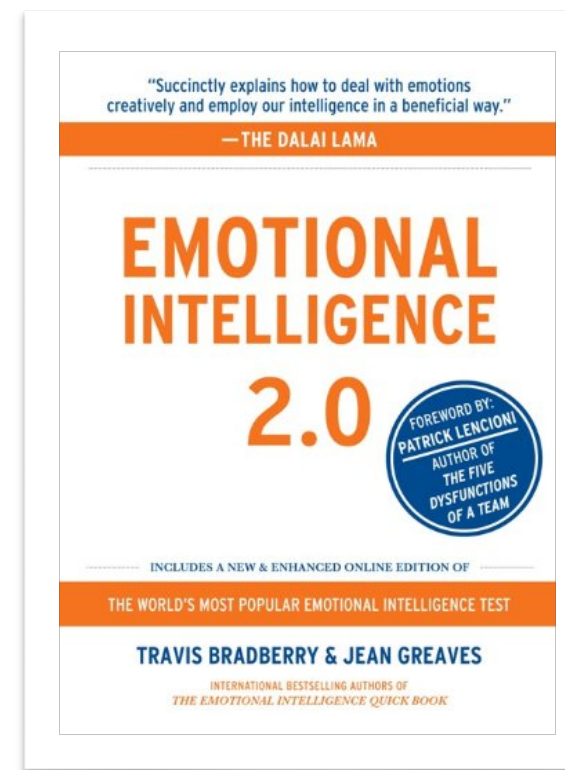
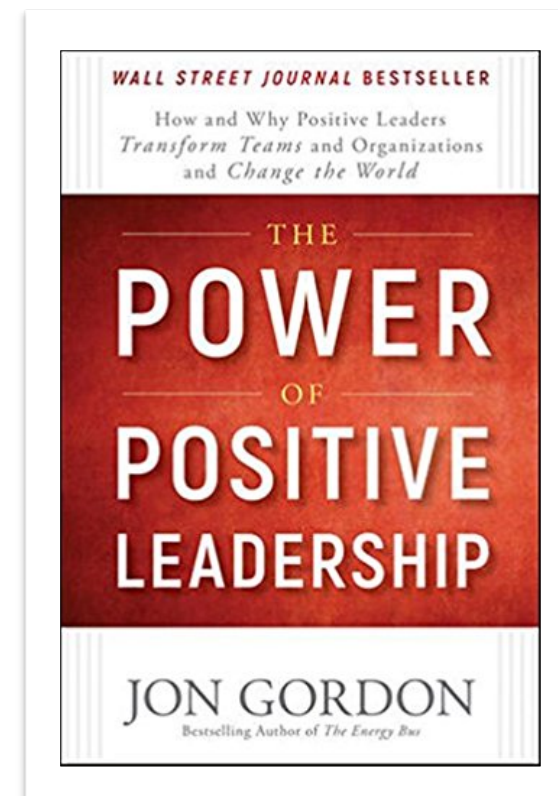
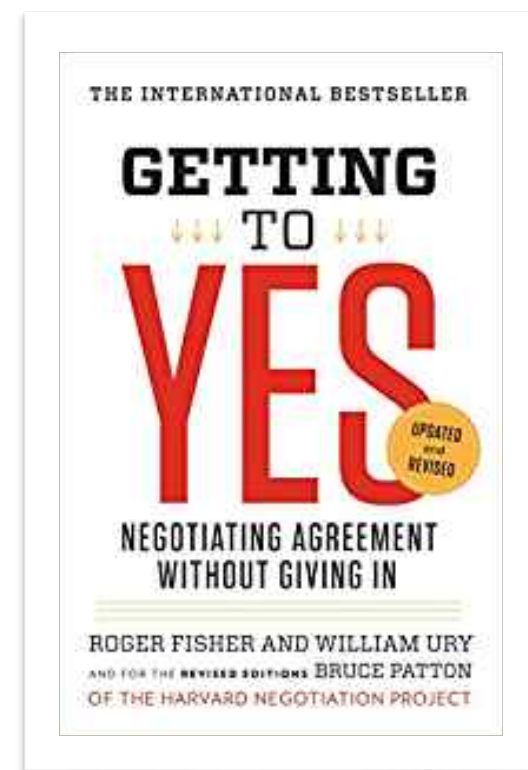
Coaching

Feedback

Influencing

Delegation

Leadership Skills



Exploring Technical Leadership



Reasons For You



A man with light hair and a beard, wearing a white button-down shirt, is shown from the chest up. He has his eyes closed and his hands pressed against his temples, suggesting a headache or intense stress. A pair of glasses is perched on his head. He is wearing a watch on his left wrist. The entire image is overlaid with a semi-transparent red filter. A white rectangular box is positioned in the lower half of the image, containing the text "I'm bored".

"I'm bored"

A man with light hair and a beard, wearing a white button-down shirt, is shown from the chest up. He has his eyes closed and his hands pressed against his temples, suggesting a headache or frustration. A watch is visible on his left wrist. The entire image is overlaid with a semi-transparent red filter. A white rectangular box is positioned in the lower half of the image, containing text.

*"How many times do I need to build
a login screen?"*

A man with blonde hair and a beard, wearing a white button-down shirt, is shown from the chest up. He has his eyes closed and his hands pressed against his temples, indicating a headache or mental anguish. In the background, a knife is visible, partially obscured by his hands. The entire image has a strong red color cast. A large white rectangular box is overlaid on the lower half of the image, containing text.

*“I write programs that read JSON,
calculate something and then write
out JSON”*

The background of the slide is a photograph of several books stacked on a shelf. The books have various colored spines, including shades of purple, blue, and brown. A semi-transparent purple overlay covers the entire image. In the upper portion, there is a white rectangular box containing the title text.

Reason 1: Build different skills

A man with light hair and a beard, wearing a white button-down shirt, is shown from the chest up. He has his eyes closed and his hands pressed against his temples, suggesting a headache or frustration. A watch is visible on his left wrist. The background is a solid red color. A white rectangular box is overlaid on the lower half of the image, containing the text.

“My tech lead doesn’t seem to do anything!”

A man with light hair and a beard, wearing a white button-down shirt, is shown from the chest up. He has his eyes closed and his hands pressed against his temples, suggesting a headache or intense stress. A watch is visible on his left wrist. The entire image is overlaid with a semi-transparent red filter. A white rectangular box is positioned in the lower half of the image, containing text.

*“Why do we have to write a
design doc for every feature?”*

A man with blonde hair and a beard, wearing a white button-down shirt, is shown from the chest up. He has his eyes closed and his mouth open in a pained expression, with both hands pressed against his temples. A knife is visible in his right hand, held near his head. He is also wearing a wristwatch on his left wrist. The entire image is overlaid with a semi-transparent red filter. A white rectangular box is positioned in the lower half of the image, containing the text.

*“I have ideas how to make our
team better”*



Reason 2: Shape the team environment

A man with light hair and a beard, wearing a white button-down shirt, is shown from the chest up. He has his eyes closed and his hands pressed against his temples, suggesting a headache or intense stress. A pair of glasses is perched on his head. He is wearing a watch on his left wrist. The entire image is overlaid with a semi-transparent red filter. A white rectangular box is positioned across the middle of the image, containing the text.

“No one’s solving that problem”

A man with light hair and a beard is shown from the chest up, wearing a white button-down shirt. He has his eyes closed and his hands pressed against his temples, indicating a headache or intense stress. On his left wrist, a gold-toned watch with a white face and black numerals is visible. In the background, a knife is positioned vertically, its handle at the top and blade pointing downwards. The entire image is overlaid with a semi-transparent red filter. A white rectangular box is centered over the lower half of the image, containing text.

*“Why aren’t we getting rid of that
old unsupported library?”*

A man with light hair and a beard, wearing a white button-down shirt, is shown from the chest up. He has his eyes closed and his mouth open in a pained expression, with both hands pressed against his temples. A pair of glasses is perched on his head. The entire image is overlaid with a semi-transparent red filter. A white rectangular box is positioned in the lower half of the image, containing the text.

*“I want to get involved earlier in
the product design”*

*Reason 3: Have greater
impact and influence*

A man with light hair and a beard is shown from the chest up, wearing a white button-down shirt. He has his eyes closed and his hands pressed against his temples, indicating a headache or intense stress. A pair of glasses is perched on his head. The entire image is overlaid with a semi-transparent red filter. A white rectangular box is positioned across the lower half of the image, containing the text.

“They’re wasting their potential”

Reason 4: See growth in others



A man with blonde hair and a beard, wearing a white button-down shirt, is shown from the chest up. He has his eyes closed and his mouth open in a pained expression, with both hands pressed against his temples. On his left wrist, a gold-toned watch with a white face and black numerals is visible. In the background, a knife is positioned vertically, its handle pointing upwards and its blade pointing downwards. The entire image is overlaid with a semi-transparent red filter. A white rectangular box is centered horizontally across the middle of the image, containing the text "This is an easy problem!".

"This is an easy problem!"



*Reason 5: Work on more
challenging problems*

Reasons For You



How to Get Started





Recipe



1. Look for misalignment/neglect

Example

“Why does everyone have their own special build script?”

Example

"Pull request reviews are really different depending on who completes it"

Example

“Why haven’t we got rid of this outdated tool?”

Recipe



2. Choose to own the issue

- 
- ▶ Patience required
 - ▶ Change takes time
 - ▶ Step by step

Recipe



3. Seek sponsorship

A lifebuoy with orange and white stripes is hanging from a yellow metal railing. The background is a body of water with a light blue tint. The railing has horizontal bars and a vertical post. The lifebuoy is made of foam and has a rope attached to it.

Your safety first

Who is your sponsor?

Recipe



4. Deliver value

Finish what you start



Recipe



5. Celebrate (and repeat)

Recipe

1. Look for misalignment/neglect
2. Choose to own the issue
3. Seek sponsorship
4. Deliver value
5. Celebrate (and repeat)



*"It's easier for managers to
recognise what you are
already doing, not what
you **might** do"*

Exploring Technical Leadership



Reasons For You



How to Get Started



Questions?

Need help training your technical leaders
and managers?

Get in touch at pat@patkua.com or
<https://www.linkedin.com/in/patkua/>

